

# Ahmed Baig

Islamabad, Pakistan

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## SUMMARY

Backend engineer with 3 years of experience designing and building scalable, high-performance production APIs with Node.js/NestJS and Python on PostgreSQL. Strong track record in relational database performance, third-party API integration (Twilio, AWS, Google Gemini), and containerized delivery (Docker, CI/CD). Hands-on experience building LLM-powered features and automation workflows, including tool/function calling, prompt engineering, and structured output handling.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, SQL, HTML/CSS

**Backend & APIs:** Node.js, NestJS, Express.js, Flask, REST API design, GraphQL, WebSockets, Swagger/OpenAPI

**API & Systems:** Idempotency, pagination, rate limiting, authentication (JWT), caching (Redis), profiling & load optimization, automated testing (Jest), observability/structured logging

**Databases:** PostgreSQL, PostGIS, Redis, MySQL, MongoDB, DynamoDB; ORMs: Sequelize, TypeORM

**AI & Automation:** LLM API integration (Google Gemini), tool/function calling, prompt engineering, structured output, n8n workflow automation

**Cloud & DevOps:** AWS (Lambda, Cognito, Amplify, DynamoDB, S3), Docker, GitHub Actions, Jenkins, Terraform

**Frontend:** React, Next.js, React Query

## EDUCATION

**National University of Computer and Emerging Sciences**

*Bachelor of Science in Computer Science*

Islamabad, Pakistan

2020 – 2024

## EXPERIENCE

**Backend Specialist**

*BaKhabar Kissan (BKK)*

June 2025 – Present

*Islamabad, Pakistan*

- Developing robust, high-performance backend systems and APIs using Node.js, NestJS, and Python to drive core business functionalities.
- Designing and implementing end-to-end database and system flow architectures on PostgreSQL, laying the foundation for scalable and maintainable applications.
- Engineering and maintaining complex ETL pipelines, ensuring reliable and efficient data extraction, transformation, and loading across multiple systems.
- Leading deep-dive optimizations of relational database performance through profiling and query tuning, specializing in advanced SQL and PostGIS (Materialized Views, spatial indexing, complex Joins).
- Architecting system-level performance enhancements by reducing time/space complexity and designing parallel execution workflows to keep systems dynamically scalable under load.
- Driving ecosystem-wide performance optimization across 9 projects; spearheading complete end-to-end backend architecture for 5 key projects and frontend delivery for 3 portals (React/Next.js).

**Full Stack Engineer**

*QLU.ai*

Aug 2024 – May 2025

*Islamabad, Pakistan*

- Designed and implemented communication systems (incoming/outgoing calls, messaging) by integrating Twilio's third-party API, including callback/webhook handling for real-time events.
- Optimized API performance and data processing, enabling efficient handling of 80,000+ data items in a single API call through algorithmic improvements and database optimizations.
- Developed and enhanced production-level features using NestJS, React, React Query, and PostgreSQL, ensuring scalable and efficient web applications.
- Built and expanded core functionalities such as anonymous contact handling, contact notes, and real-time messaging, significantly improving user experience and workflow efficiency.
- Developed a **real-time chat application** using WebSockets and the PERN stack, ensuring low-latency communication and robust performance.

**Fellowship Associate - Backend**

*Fasset*

June 2024 – Aug 2024

*Islamabad, Pakistan*

- Contributed to developing a **Flexible Transaction Fee System**, implementing fee calculation logic, JWT authentication, Redis caching, and input validation using Sequelize ORM (PostgreSQL).
- Gained expertise in MVC backend architecture while collaborating with Fasset's national and international teams during the fellowship.

#### Teaching Assistant (CS4032: Web Programming)

Feb 2024 – Jun 2024

National University of Computer and Emerging Sciences

Islamabad, Pakistan

- Conducted interactive demo sessions on core web development concepts, including JavaScript, React.js, and Node.js, for a class of 60 students.

#### Backend Development Intern

July 2023 – Aug 2023

MRS Technologies

Islamabad, Pakistan

- Built the backend of an **IoT-based microgreens plant system** using AWS Amplify, DynamoDB, GraphQL, and AWS Lambda with Cognito sign-up. Selected as one of the top 15 interns out of 13,000 candidates.

#### Web Development Intern

Aug 2022 – Sept 2022

Musketeers Tech (Private) Ltd

Lahore, Pakistan

- Contributed to a **Pet Store Management System** using the MERN stack, Docker, and Git/GitHub.

## PROJECTS

#### AI Financial Analyst – LLM Automation Pipeline | *n8n, Docker, Google Gemini, JavaScript, REST APIs*

- Built an automated daily market-analysis pipeline on self-hosted n8n (Docker): tool nodes call external APIs to ingest and normalize global RSS feeds, then invoke the Google Gemini LLM to synthesize macroeconomic trends into structured output.
- Applied prompt engineering for consistent structured summaries with scheduled, automated delivery to a downstream channel.

#### Centralized Auth | *NestJS, TypeORM, PostgreSQL, JWT, Redis, Swagger, Jest, Docker*

- Built a centralized auth system with JWT token rotation, dashboard-scoped RBAC, multi-tenant partner isolation, API key management, and full audit logging.
- Implemented 65 automated E2E tests with database-backed HTML reporting, plus rate limiting and request correlation tracing for observability.

#### Sugarcane Monitoring Dashboard | *NestJS, TypeORM, PostgreSQL, PostGIS, Next.js, GeoServer, JWT*

- Built a multi-service geospatial platform (NestJS mill backend + field survey service) with PostGIS spatial queries for boundary analysis, overlap detection, and harvest tracking, serving 60K+ crop samples with disease/pest classification.
- Implemented RBAC with multi-tenant mill isolation, S3-backed field survey image uploads with EXIF geotag extraction, Excel/PDF report generation, and interactive WMS dashboards (Next.js).

#### Remote Crop Monitoring Dashboard | *NestJS, TypeORM, PostgreSQL, PostGIS, Next.js, GeoServer, JWT*

- Developed a NestJS backend integrating GeoServer, NDVI satellite imagery, and real-time third-party weather APIs for precision agriculture, with automated E2E tests covering the API pipeline.
- Implemented JWT auth, paginated access to 3,000+ farmer records, and an interactive Next.js dashboard with WMS/WFS map layers and farm health scoring.

#### Dialer – VoIP Communication Platform | *NestJS, PostgreSQL, Sequelize, Twilio, Socket.io, React*

- Developed a full-stack VoIP platform integrating Twilio's third-party APIs for real-time calling, messaging, voicemail, and CSV contact upload.
- Integrated AI-powered transcription endpoints for real-time speech-to-text and built scalable APIs with NestJS and PostgreSQL.

#### Flexible Transaction Fee System | *Node.js, PostgreSQL, Sequelize, JWT*

- Engineered a dynamic transaction fee calculation engine (jurisdictions, transaction types, user segments) with threshold waivers, activity-based tiered discounts, and JWT-secured administrative APIs.

## CERTIFICATIONS

|                                                             |      |
|-------------------------------------------------------------|------|
| Fundamentals of Remote Sensing (NASA)                       | 2025 |
| AWS Academy Microservices and CI/CD Pipeline Builders (AWS) | 2024 |
| freeCodeCamp: Back End Development and APIs                 | 2023 |
| freeCodeCamp: JavaScript Algorithms and Data Structures     | 2022 |
| DataCamp: Introduction to Python                            | 2022 |